

Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.

Key Characteristics of the Dataset:

- **Shape:** 167,826 rows x 9 columns
- **Columns:**
 - Country/Region, UNICEF_Region, Indicator, Year, Sex, Age, Value, Lower, Upper
- **Data Types:**
 - Value, Lower, and Upper are stored as **strings**, which will likely need to be converted to numerical types for analysis.
 - Year is a float but should ideally be an integer.
- **Missing Data:**
 - UNICEF_Region : 10,432 missing values
 - Lower & Upper : 57 missing values each

Let me know the specific analytical question you'd like answered using this dataset (e.g., trends, comparisons, aggregations), and I'll provide a focused analysis and the Python code used. [↕]



Propose three interventions that integrate demographic targeting and regional data to reduce adolescent HIV mortality. Rank these interventions based on impact using measurable criteria such as cost-effectiveness, feasibility, and scalability.

To propose and rank interventions effectively, I will:

1. **Analyze demographic and regional trends** in adolescent HIV mortality using the dataset.
2. **Identify high-risk groups and regions** based on measurable data.
3. **Propose three interventions**, integrating these insights.
4. **Rank them** using a scoring system for:
 - Cost-effectiveness
 - Feasibility
 - Scalability

🔍 Step 1: Extract Relevant Data (Adolescent HIV Mortality by Region, Sex, Age)

Let's isolate data where the indicator is related to **HIV-related deaths** among adolescents.

The data shows that **Eastern and Southern Africa** and **West and Central Africa** have the highest adolescent HIV mortality rates, with both **male** and **female** adolescents heavily affected.

✅ Proposed Interventions (with Target ↕)

- **Target:** Adolescents aged 10–19 in **Eastern and Southern Africa**
- **Demographic Insight:** High mortality across both sexes
- **Why:** Schools are high-density, high-access environments

Ranking:

Metric	Score (1–5)	Justification
Cost-Effectiveness	4	Leverages existing infrastructure
Feasibility	5	High school enrollment in region
Scalability	5	Can expand across similar countries

2. Gender-Specific Mobile Clinics for HIV Care

- **Target:** Female adolescents in rural areas of **West and Central Africa**
- **Demographic Insight:** Females in this region show particularly high death rates

Ranking:

Metric	Score (1–5)	Justification
Cost-Effectiveness	3	Operational costs of mobile clinics
Feasibility	4	Proven models exist but need adaptation
Scalability	3	Logistics depend on road access & funding

3. Digital Peer-Support and ART Adherence Apps

- **Target:** Male adolescents aged 15–19 in both regions
- **Demographic Insight:** High male mortality possibly due to late care-seeking

Ranking:

Metric	Score (1–5)	Justification
Cost-Effectiveness	5	Low marginal cost after setup
Feasibility	3	Mobile penetration growing but uneven
Scalability	5	Easily deployable in multiple regions

 Final Intervention Ranking

1. **School-Based HIV Testing and Treatment**
2. **Digital Peer-Support Apps**
3. **Gender-Specific Mobile Clinics**

Would you like the full scoring table in code form, or visualized as a chart? [\[↗\]](#)

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O ChatGPT pode cometer erros. Considere verificar informações importantes.